Manikanda Prabhu Bagavathi Krishnan

Data Analyst | Machine Learning Engineer
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PROFESSIONAL SUMMARY

Data analyst and machine learning engineer finishing a Master of Data Science at RMIT University in December 2026, with close to two years of hands-on SQL and Python work. At Smartfusion I analysed 1M+ operational records, built Python pipelines that lifted reporting accuracy by 20% and set up the data models behind Power BI and Tableau dashboards for five business units. At Focus Bear I built a wearable breathing and posture tracker with the CEO and a multi-model AI classifier that reached 94.7% accuracy, and on my RMIT capstone I led data processing for an AI coursebook checking pipeline backed by 122 automated tests. I care most about data people can trust, and I enjoy explaining it to people who don't work with it every day.

Right to work: Student Visa (subclass 500), moving to a Temporary Graduate Visa (subclass 485) with full work rights. No sponsorship required.

TECHNICAL SKILLS

Programming: Python, SQL, R, JavaScript

Data Analysis: Statistical Analysis, Exploratory Data Analysis (EDA), Data Modeling, ETL, Data Cleaning, Data Validation, Data Quality, Trend Analysis, Reporting

BI and Visualisation: Power BI, Tableau, Excel, Dashboards, KPI Reporting

Machine Learning and AI: Machine Learning, Pandas, NumPy, Scikit-learn, TensorFlow, LLM Evaluation (GPT, Gemini, Claude), Prompt Engineering, Ensemble Classification, NLP, SymPy

Tools: Git and GitHub, pytest, Regression Testing, Jupyter, Streamlit, MongoDB

Hardware and Signals: Signal Processing, ESP32-S3 Firmware, IMU Sensor Validation, Bluetooth LE and USB Serial Streaming

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern | Focus Bear, Melbourne VIC **Apr 2026 to Present**

Breathing and Posture Detection, Wearable MVP (Jul 2026 to Present)

- Delivered a working end-to-end dual-IMU breathing and posture detection MVP in direct collaboration with the CEO, from M5Stack StickS3 hardware through a Python signal processing engine to a macOS menu-bar app
- Diagnosed and fixed a posture calibration bug where the upright baseline was captured from an arbitrary sensor position, and replaced it with a user-triggered calibration flow verified on real hardware
- Resolved a firmware flashing failure on ESP32-S3 hardware caused by an incompatible reset handshake, using an alternative esptool-based procedure to unblock hardware deployment
- Built and iterated on a macOS menu-bar app (Python, rumps, AppKit), redesigning the calibration screen after CEO feedback into a single instruction and button
- Validated dual-IMU (BMI270) breathing signal accuracy through structured hardware testing over USB serial and Bluetooth LE, confirming correct chest and belly breathing differentiation

Attention Score Prediction, Vision-Language Model Classifier (Apr 2026 to Jun 2026)

- Achieved 94.7% consensus accuracy on a 58-frame labelled test set by engineering a multi-model ensemble classifier using GPT-5.4 and Gemini 3.1 Flash Lite with Claude Sonnet as tiebreaker, outperforming each individual model at 91.4%
- Improved category accuracy from 72% to 91.4% across 6 automated prompt-tuning iterations, replacing manual trial and error
- Engineered separate prompt architectures for GPT and Gemini vision models to classify user attention states from screen and workspace images, feeding one consensus classifier
- Built a label normalisation step that flagged and corrected invalid model responses before scoring, and a consensus pipeline with a 98.3% agreement rate that flags conflict frames for manual review

Data Processing and ML Lead | Spectrum Tuition (RMIT Capstone), Melbourne VIC **Aug 2026 to Nov 2026**

Automated Curriculum Processing Engine, industry client project with a team of 5

- Designed and built a routed answer validator in Python and JavaScript that uses SymPy and exact rational arithmetic to check coursebook answers across 7 question types, with three outcomes (verified, mismatch, needs human review) so uncertain cases are never guessed
- Verified 192 of 1,000 blocks on a real Year 8 Maths Tutor Book and 109 of 4,155 on a Year 4 book with zero false mismatches, by finding and fixing defects with regression testing (71 Python and 51 JavaScript tests)
- Built a browser-based validation tool producing CSV and JSON reports, and demonstrated it to the client in weekly sprint reviews, adopting their feedback on wording and routing rules
- Supported the LLM checking step with a reusable Gemini client and a curriculum-tagging prompt, and documented extraction gaps with block-level evidence for the pipeline team

- Reviewed pull requests by running branches, followed Git workflow standards, and wrote handover notes and a layout-check report for the client on Grade 4 books

Data Analyst | Smartfusion Corporate Solutions, Chennai, India **Feb 2024 to Feb 2025**

- Analysed 1M+ operational records using Python and SQL, applying statistical analysis including significance tests to identify trends and bottlenecks across five business units, supporting a process redesign that reduced manual processing overhead by 30%
- Built automated ETL data validation and cleaning pipelines in Python, improving reporting accuracy by 20% and strengthening data quality across operational dashboards
- Handled the data modeling behind interactive Power BI and Tableau dashboards, setting up the tables and relationships, so five business units could monitor real time KPIs with fewer decision making delays

Client Partner | Access Healthcare, Chennai, India **Nov 2021 to Apr 2023**

- Analysed large scale healthcare claims datasets to identify approval bottlenecks, and implemented structured tracking workflows that increased claim approval rates by 40%

PROJECTS

Credit Card Fraud Detection Using GANs Python, TensorFlow, Scikit-learn, Pandas

- Engineered a GAN based synthetic data augmentation pipeline for anomaly detection on a 284K transaction dataset with a 0.17% fraud rate, achieving F1 = 0.805 and 99.94% accuracy, outperforming SMOTE by 11.3% precision

InfoMate, AI Powered University Information Chatbot Python, GPT-4o, Streamlit, MongoDB

- Built an NLP chatbot on 10,000+ structured and unstructured documents using GPT-4o embeddings, validated with 10 domain expert faculty members, and deployed as a Streamlit app with MongoDB backed session management

EDUCATION

Master of Data Science | RMIT University, Melbourne **Feb 2025 to Dec 2026**

GPA 3.4/4.0. Coursework: Database Concepts, Big Data Management, Data Visualisation, Practical Data Science, Machine Learning, Time Series Analysis

Bachelor of Technology, Information Technology | Jeppiaar Institute of Technology, India **Aug 2016 to Nov 2020**

GPA 3.12/4.0. Majors: Computer Systems, Information Systems

CERTIFICATIONS

- Deloitte Australia, Data Analytics Job Simulation (Forage): built data visualisations in Tableau and applied data analysis to business decision making scenarios
- Commonwealth Bank, Introduction to Data Science Job Simulation (Forage): developed data analysis and visualisation solutions for financial decision making contexts

VOLUNTEER EXPERIENCE

Event Volunteer | PMI Melbourne Chapter **2025 to Present**

- Coordinated check-in, setup and speakers and ran a session across 2 community events of 60+ attendees, and mentored new and existing members

Event Crew | RMIT University Open Day **2025**

- Guided prospective students through campus and program information sessions